# JK Flip Flops

## jk_ff_all_named.v

```
`timescale 1ns/1ps

module jk_ff_all_named(
    input clk,
    input rst,
    input set,
    input en,
    input j,
    input k,

    output reg q_basic,
    output reg q_ifelse,
    output reg q_async_reset,
    output reg q_async_set,
    output reg q_enable,
    output reg q_equation,
    output reg q_toggle,
    output reg q_full
);

wire d_equation;

assign d_equation = (j & ~q_equation) | (~k & q_equation);


always @(posedge clk)
begin
    case({j,k})
        2'b00: q_basic <= q_basic;
        2'b01: q_basic <= 1'b0;
        2'b10: q_basic <= 1'b1;
        2'b11: q_basic <= ~q_basic;
    endcase
```

```
end


always @(posedge clk)
begin
    if(j==0 && k==0) q_ifelse <= q_ifelse;
    else if(j==0 && k==1) q_ifelse <= 0;
    else if(j==1 && k==0) q_ifelse <= 1;
    else q_ifelse <= ~q_ifelse;
end


always @(posedge clk or posedge rst)
begin
    if(rst) q_async_reset <= 0;
    else begin
        case({j,k})
            2'b00: q_async_reset <= q_async_reset;
            2'b01: q_async_reset <= 0;
            2'b10: q_async_reset <= 1;
            2'b11: q_async_reset <= ~q_async_reset;
        endcase
    end
end


always @(posedge clk or posedge set)
begin
    if(set) q_async_set <= 1;
    else begin
        case({j,k})
            2'b00: q_async_set <= q_async_set;
            2'b01: q_async_set <= 0;
            2'b10: q_async_set <= 1;
            2'b11: q_async_set <= ~q_async_set;
        endcase
    end
end
```

```
always @(posedge clk)
begin
    if(en) begin
        case({j,k})
            2'b00: q_enable <= q_enable;
            2'b01: q_enable <= 0;
            2'b10: q_enable <= 1;
            2'b11: q_enable <= ~q_enable;
        endcase
    end
end


always @(posedge clk)
    q_equation <= d_equation;


always @(posedge clk)
begin
    if(j==0 && k==0) q_toggle <= q_toggle;
    else if(j==1 && k==1) q_toggle <= ~q_toggle;
    else if(j==1) q_toggle <= 1;
    else q_toggle <= 0;
end


always @(posedge clk or posedge rst or posedge set)
begin
    if(rst) q_full <= 0;
    else if(set) q_full <= 1;
    else if(en) begin
        case({j,k})
            2'b00: q_full <= q_full;
            2'b01: q_full <= 0;
            2'b10: q_full <= 1;
            2'b11: q_full <= ~q_full;
```

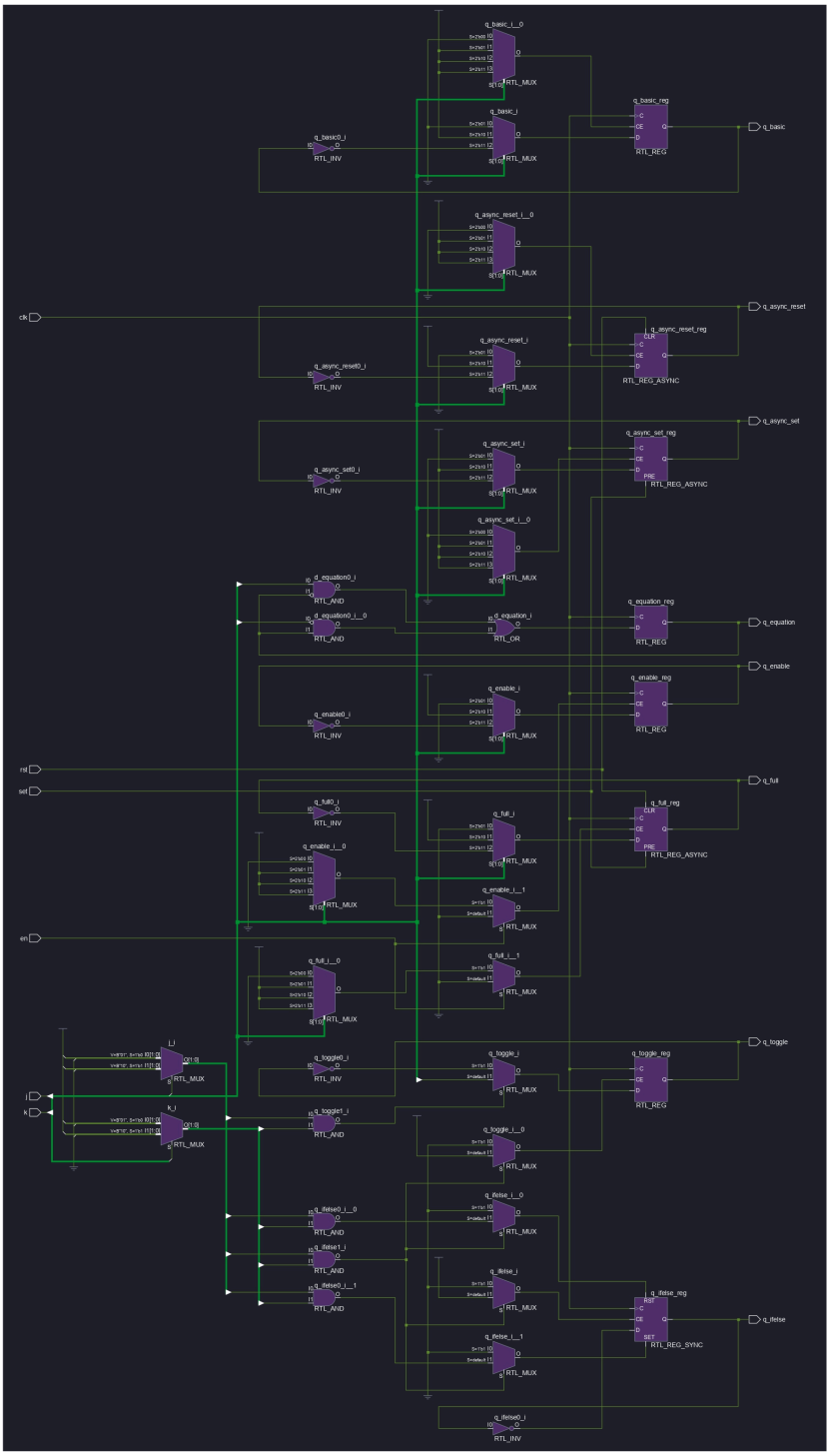
```
        endcase
    end
end

endmodule
```

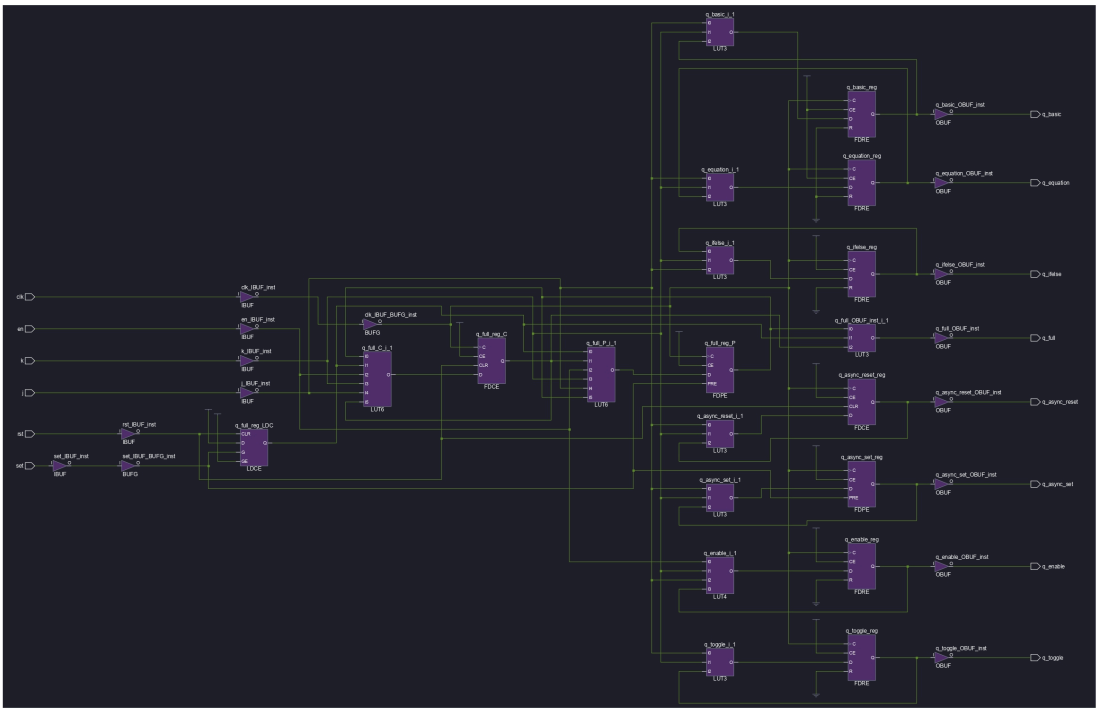
## jk_ff_all_named_tb.v

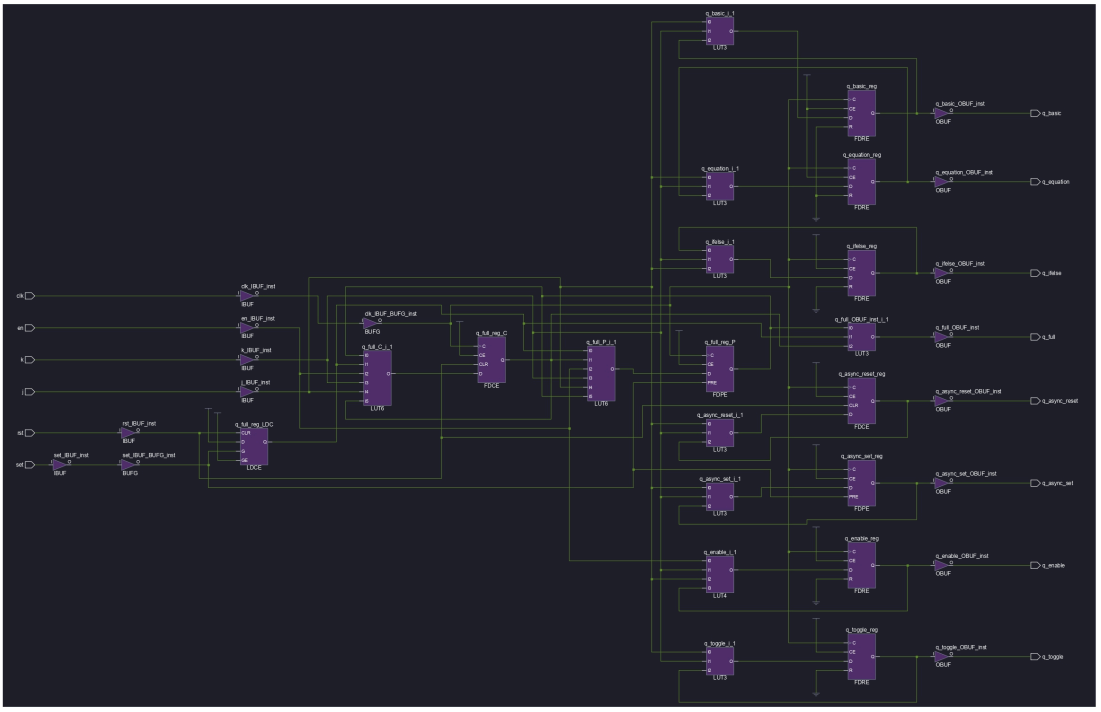
```
`timescale 1ns/1ps

module jk_ff_all_named_tb;

reg clk;
reg rst;
reg set;
reg en;
reg j;
reg k;

wire q_basic;
wire q_ifelse;
wire q_async_reset;
wire q_async_set;
wire q_enable;
wire q_equation;
wire q_toggle;
wire q_full;

jk_ff_all_named uut (
    .clk(clk),
    .rst(rst),
    .set(set),
    .en(en),
    .j(j),
    .k(k),
```

```
    .q_basic(q_basic),
    .q_ifelse(q_ifelse),
    .q_async_reset(q_async_reset),
    .q_async_set(q_async_set),
    .q_enable(q_enable),
    .q_equation(q_equation),
    .q_toggle(q_toggle),
    .q_full(q_full)
);

initial clk = 0;
always #5 clk = ~clk;

initial begin

    rst = 0; set = 0; en = 1;
    j = 0; k = 0;

    #10 j=0; k=0;
    #10 j=0; k=1;
    #10 j=1; k=0;
    #10 j=1; k=1;

    #10 rst = 1;
    #10 rst = 0;

    #10 set = 1;
    #10 set = 0;

    #10 en = 0;
    #20 en = 1;

    #10 j=1; k=1;
    #10 j=0; k=1;
    #10 j=1; k=0;

    #50 $finish;
```

```
end

endmodule
```




ck
en
k
j
rst
set
ck_IBUF_inst
IBUF
en_IBUF_inst
IBUF
k_IBUF_inst
IBUF
j_IBUF_inst
IBUF
rst_IBUF_inst
IBUF
set_IBUF_inst
IBUF
set_IBUF_BUFG_inst
BUFG
ck_IBUF_BUFG_inst
BUFG
q_full_C_i_1
LUT6
q_full_reg_C
FDCE
q_full_reg_LDC
LDCE
q_full_P_i_1
LUT6
q_basic_i_1
LUT3
q_basic_reg
FDRE
q_basic_OBUF_inst
OBUF
q_basic
q_equation_i_1
LUT3
q_equation_reg
FDRE
q_equation_OBUF_inst
OBUF
q_equation
q_false_i_1
LUT3
q_false_reg
FDRE
q_false_OBUF_inst
OBUF
q_false
q_full_OBUF_inst_i_1
LUT3
q_full_OBUF_inst
OBUF
q_full
q_full_reg_P
FDPE
q_async_reset_reg
FDCE
q_async_reset_OBUF_inst
OBUF
q_async_reset
q_async_reset_i_1
LUT3
q_async_set_i_1
LUT3
q_async_set_reg
FDPE
q_async_set_OBUF_inst
OBUF
q_async_set
q_enable_i_1
LUT4
q_enable_reg
FDRE
q_enable_OBUF_inst
OBUF
q_enable
q_toggle_i_1
LUT3
q_toggle_reg
FDRE
q_toggle_OBUF_inst
OBUF
q_toggle


q_basic_i_1
LUT3
q_basic_reg
FDRE
q_basic_OBUF_inst
OBUF
q_basic
q_equation_i_1
LUT3
q_equation_reg
FDRE
q_equation_OBUF_inst
OBUF
q_equation
q_false_i_1
LUT3
q_false_reg
FDRE
q_false_OBUF_inst
OBUF
q_false
q_full_OBUF_inst_i_1
LUT3
q_full_OBUF_inst
OBUF
q_full
q_async_reset_reg
FDCE
q_async_reset_OBUF_inst
OBUF
q_async_reset
q_async_set_reg
FDPE
q_async_set_OBUF_inst
OBUF
q_async_set
q_enable_reg
FDRE
q_enable_OBUF_inst
OBUF
q_enable
q_toggle_reg
FDRE
q_toggle_OBUF_inst
OBUF
q_toggle
ck
ak
k
l
ni
set
ck_IBUF_inst
IBUF
ak_IBUF_inst
IBUF
k_IBUF_inst
IBUF
l_IBUF_inst
IBUF
ni_IBUF_inst
IBUF
set_IBUF_inst
IBUF
set_IBUF_BUFG_inst
BUFG
ck_IBUF_BUFG_inst
BUFG
q_full_C_i_1
LUT6
q_full_reg_C
FDCE
q_full_P_i_1
LUT6
q_full_reg_LDC
LDCE
q_full_reg_P
FDPE
q_async_reset_i_1
LUT3
q_async_set_i_1
LUT3
q_enable_i_1
LUT4
q_toggle_i_1
LUT3